# A Centralized System for Real-Time Farm Animal Disease Reporting

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Project Proposal Report

Kumarasinghe S.S – IT22221414

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Computer Science & Software Engineering
Sri Lanka Institute of Information Technology Sri Lanka

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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

| Name | Student ID | Signature |
|---|---|---|
| Kumarasinghe S.S | IT22221414 | |

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

| Name | Date | Signature |
|---|---|---|
| Ms.Dilani Lunugalage | 2026 / 03 / 14 | |
| Mr. Eishan Weerasinghe | 2026 / 03 / 14 | |

# ABSTRACT

Livestock diseases are known to be a major threat to agricultural productivity and economic stability, especially in developing countries such as Sri Lanka, where livestock plays an important role in rural livelihoods. The traditional approach to veterinary surveillance is based on retrospective reporting of animal diseases, which has often shown to be ineffective in responding to and preventing animal diseases in a proactive manner.

Moreover, livestock disease records in Sri Lanka are typically sparse and irregular, which are mostly represented as long stretches of zero data and occasionally some reported outbreak data. Such zero-heavy data is usually difficult to forecast using traditional methods and deep learning algorithms. Most of the available literature on animal disease prediction has shown little use of environmental factors and has not made any attempt to provide interpretable predictions.

This research aims to develop a Climate-Informed Seasonal Livestock Disease Forecasting and Early Warning System, which can utilize data from epidemiological surveillance and environmental/climatic data to enhance the accuracy of disease outbreak prediction. The system can utilize environmental data such as rainfall, temperature, humidity, vegetation index, soil moisture, and cattle density to estimate disease risk.

A two-stage machine learning framework is employed, where the first model predicts outbreak probability and the second estimates outbreak severity. Multiple candidate algorithms can be experimented with and compared with statistical methods using quantitative metrics, such as Zero-Inflated Negative Binomial Regression. To enhance model interpretability and provide better decision-making capabilities, the model can be extended using SHapley Additive exPlanations (SHAP) and prediction intervals using Bootstrap methods.

The expected outcome of the research would be a is a climate-informed forecasting framework that would be able to offer a prediction of risks of outbreaks at the district level. It would be able to offer useful insights to veterinary authorities in the prevention of diseases, which would in turn reduce the loss of livestock and promote the sustainability of agriculture.

**Keywords:** Livestock Disease Forecasting, Machine Learning, Climate-Informed Prediction, Early Warning Systems, Veterinary Epidemiology

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# LIST OF ABBREVIATIONS

| Abbreviation | Full Form |
|---|---|
| ADRS | Animal Disease Reporting System |
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| BCS | Body Condition Score |
| B2B | Business-to-Business |
| B2G | Business-to-Government |
| CHIRPS | Climate Hazards Group InfraRed Precipitation with Station Data |
| CoT | Chain of Thought |
| CV | Computer Vision |
| DAPH | Department of Animal Production and Health |
| FMD | Foot-and-Mouth Disease |
| LSD | Lumpy Skin Disease |
| ML | Machine Learning |
| NDVI | Normalized Difference Vegetation Index |
| NASA POWER | NASA Prediction of Worldwide Energy Resources |
| OOD | Out-of-Distribution |
| RF | Random Forest |
| SHAP | SHapley Additive Explanations |
| SMAP | Soil Moisture Active Passive |
| SaaS | Software as a Service |
| SDG | Sustainable Development Goal |
| WBS | Work Breakdown Structure |
| XAI | Explainable Artificial Intelligence |
| ROC-AUC | Receiver Operating Characteristic – Area Under Curve |
| PR-AUC | Precision–Recall Area Under the Curve |
| F1-score | Harmonic Mean of Precision and Recall |

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# 1. INTRODUCTION

## 1.1 Background and Context

Livestock production is an important factor in the world food security, livelihoods of the rural areas, and agricultural economies. Cattle and other animals give some basic products like milk, meat and draft power which sustains millions of farmers all over the world. The livestock production is a key contributor to the national agricultural sector as well as households' income in Sri Lanka rural areas. Nonetheless, livestock infections like FMD and LSD are still posing a threat to the production and survival of livestock and farmers. Economic losses may be devastating due to death of livestock, decreased milk and meat production, high expenditure in veterinary services and because of the limitation of animal trade. Therefore, disease monitoring and detection of early outbreaks are critical in safeguarding livestock health and sustaining agriculture.

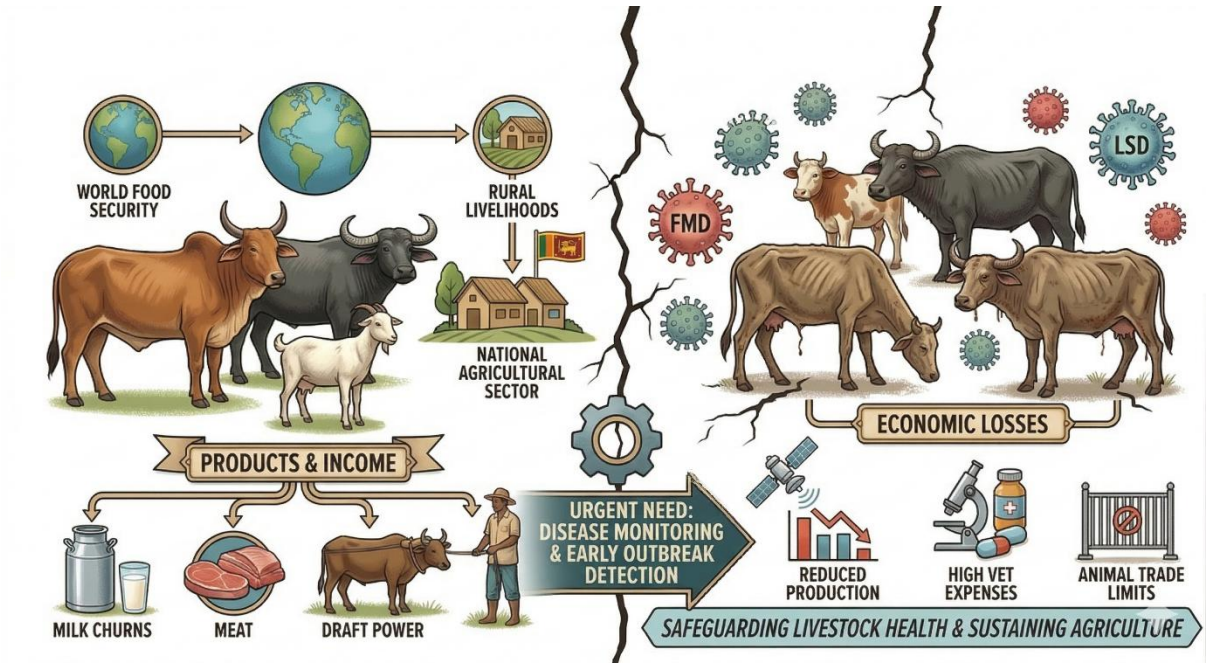



*Figure 1: Socio-Economic Impact and Pathogenic Threats to the Livestock Value Chain*

Although livestock disease surveillance is very crucial, the problem with veterinary surveillance systems in most developing countries is that they heavily depend on the retrospective reporting of disease outbreaks. In Sri Lanka, the report of disease is usually done via veterinary epidemiology bulletin which is issued by the department of animal production and health (DAPH). These reports usually have sparse and totalized outbreak information with cases of the disease reported irregularly, and with a lot of gaps in recorded outbreaks with no cases reported. These thin epidemiological data formats are problematic to traditional statistical forecasting algorithms, as well as modern deep learning models, which tend to need extensive and continuous data sets to generate accurate forecasts.

Current methods of livestock disease forecasting usually involve using past cases, or simple time-series forecasts, which can lack the ability to elucidate the compound interactions among environmental state and disease occurrence. Nevertheless, environmental and climatic conditions such as rainfall, temperature, humidity, wind speed, vegetation cover, and soil moisture have a great impact on disease outbreaks in livestock. Such environmental factors alter the effects of vectors, survival of pathogens

and animal movement, thus enhancing the risk of disease transmission in favorable conditions. Despite this association, most of the available prediction models fail to incorporate multi-source environmental information in livestock diseases prognostication models comprehensively.

To overcome the above constraints, this study recommends a Climate-Informed Seasonal Livestock Disease Forecasting and Early Warning System that incorporates epidemiological surveillance information with environmental and climatic drivers to enhance the prediction of outbreaks. The suggested solution builds a district-month dataset out of sparse disease reports and applies environmental variables on it, including rainfall, temperature, humidity, vegetation index, wind speed, soil moisture, and cattle density. A two-stage machine learning model is used to estimate the probability of occurrence of outbreaks and predict the level of outbreaks that may take place. The system seeks to deliver proactive risk predictions of outbreaks that are informed by climatic conditions and machine learning models to assist veterinary authorities in their planning of preventive interventions and effective resource allocation.

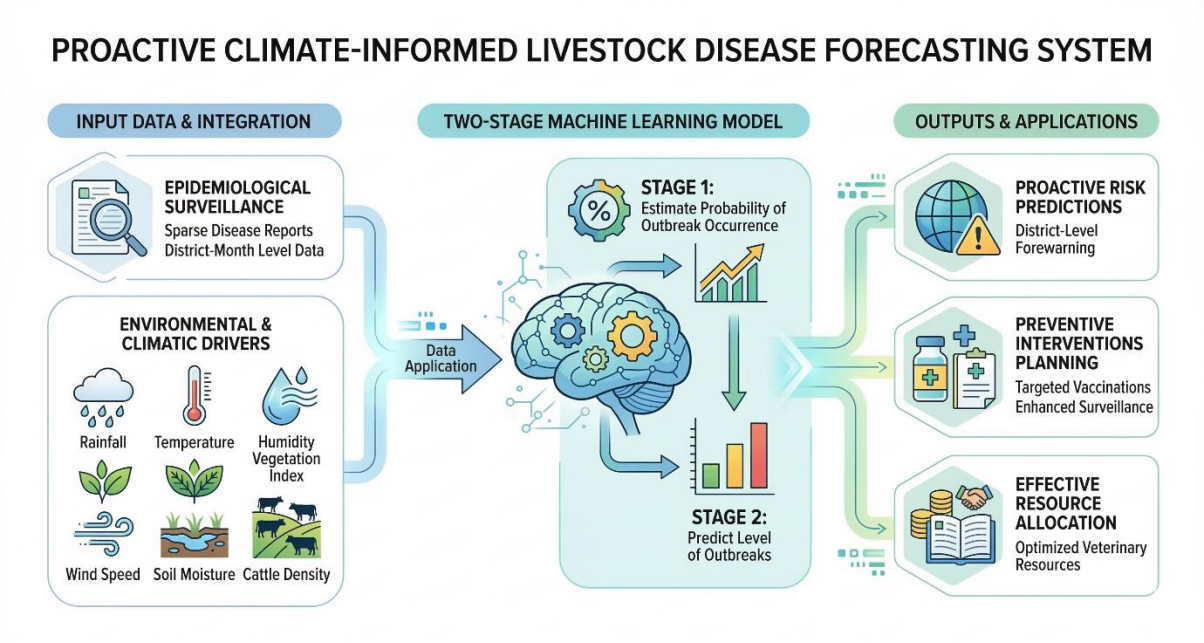



*Figure 2: Proactive climate-informed livestock disease forecasting system*

The study is consistent with the United Nations Sustainable Development Goal (SDG) 2: Zero Hunger aimed at enhancing productivity in agriculture and sustainable food systems. The proposed system helps in protecting the health of livestock by assisting in early detection and prevention of diseases in livestock, enhancing stability in food production, and enhancing the economy of the rural setting. Moreover, the research project falls within the Artificial Intelligence and Data Science research cluster, since it uses the machine learning methods, environmental data fusion, and predictive modelling to solve practical agricultural and veterinary health problems.

## 1.2 Literature Review

The diseases of livestock remain a significant threat to the global food security, sustainability of agriculture, and livelihood of rural population. Other diseases like the Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) may lead to gross economic losses because of animal losses, decreased milk and meat production, high rates of veterinary care and the limitation of livestock trade. Consequently, a lot of research has been done to come up with computational tools that would predict the epidemics of a disease and enhance the surveillance systems. These methods are not only classical statistical methods of outbreak detection but also the latest machine learning based prediction systems. Despite the importance of these studies on the dynamics of the disease, the methodology has several limitations, especially on the situation involving scanty epidemiological data, as well as environmental forces on disease transmission.

Initial studies in disease surveillance were mostly based on statistical outbreak detection algorithms that examine historical values of the number of cases to compare them with the normal increase in disease rates. Among the most popular methods is the statistical outbreak detection algorithm suggested by Farrington et al. that is based on generalized linear models, detecting the violations of the expected disease incidence patterns [1]. This has been extensively used in epidemiological surveillance systems since it can be used to notice sudden changes in cases of the disease. Equally, the Early Aberration Reporting System (EARS) was established as a surveillance system of the public health that tracks the reports of diseases and identifies abnormal trends based on statistical thresholds [2]. The methods are well-grounded in outbreak detection and are still applied in most surveillance systems. Nevertheless, they are largely reliant on regular historical reporting and rather dense time-series data.

Several studies have compared the existing classical outbreak detection algorithm performance. Comparison between various statistical detection methods have demonstrated that such methods can be effective in detecting sudden spikes in the incidence of disease in case adequate historical information is available [3]. However, their performance is lower in situations when the dataset has an irregular reporting period or long durations of zero observations. This is especially relevant in veterinary epidemiology in which the report of diseases may be infrequent and non-uniform because outbreaks are discrete and because of limitations in the surveillance infrastructure.

To overcome the issue of low disease prevalence observations, scientists have developed statistical models that are specialized in dealing with zero-inflated epidemiological data. The use of zero-inflated and hurdle regression models has been prevalent in disease modelling where the number of zero observations are compared to what is expected in disease counts models. As an illustration, observations of infectious disease data models based on zero-inflated and hurdle models have shown that the isolation of the process of disease occurrence and the process of disease severity can greatly enhance the level of prediction [4]. Comparative analyses of the zero-inflated and the hurdle models also reveal that the two models offer a more realistic representation of the disease dynamics in cases where there are intermittent outbreaks

and excess zeros in the data [5]. Such results point to the significance of the explicit modeling of the prevalence of disease outbreaks independent of the severity of the outbreaks.

Livestock disease prediction has also been done using such statistical methods. An outbreak prediction study of province-level outbreaks of the Foot-and-Mouth Disease used a zero-inflated negative binomial (ZINB) model to overcome the high rate of districts with no outbreaks in most periods of observation [6]. The analysis revealed that the use of zero-inflated modelling structures enhanced the use of disease occurrences patterns on livestock surveillance information. Although these models are suitable in the representation of zero-heavy distribution, they make use of history of the number of diseases as a source of information, and they tend to ignore environmental aspects which contribute to the spread of diseases.

New studies have focused more on the significance of environmental and climatic factors in the dynamics of livestock diseases. Climatic conditions that may have effects on the survival of pathogens, the population of vectors, and even the movements of livestock may include rainfall, temperature, humidity, and the cover of vegetation. Research with long term livestock disease patterns as examined with periodic regression models has revealed that most livestock diseases possess strong seasonal patterns which are dependent on the availability of environmental conditions [7]. These results indicate that the environmental drivers can greatly enhance the predictive models by increasing the level of accuracy of the disease forecasting systems.

It is based on this insight that several studies have used machine learning methods to predict animal diseases. Machine learning models can model complicated nonlinear interactions amid the environmental factors and occurrence of the disease. Indicatively, a study that forecasted Lumpy Skin Disease outbreaks through the predictive meteorological and geospatial variables established that environmental factors like precipitation, temperature, vapor pressure and vegetation indices greatly determine the risk of the disease [8]. The effectiveness of the random forest model was found to be especially effective in predicting nonlinear interactions of the environmental factors and prevalence of the disease.

The Ensemble machine learning methods have also been used to predict livestock disease outbreaks in other studies. Ensemble models that have integrated algorithms like random forest, gradient boosting, and AdaBoost have been used to study outbreaks of Lumpy Skin Disease in various geographical locations [9]. The models had high predictive performance and showed the relevance of environmental and geographical characteristics in the disease prediction systems. Nevertheless, most of these studies are largely binary classification of disease prevalence and most of them are based using balanced datasets that are not fully representative of the sparse and non-uniform nature of real-world surveillance data.

Besides predictive accuracy, the research has also given weight to interpretability in machine learning-based disease prediction systems in recent years. Explainable artificial intelligence (XAI) methods like SHapley Additive exPlanations (SHAP) have been established to enhance the understandability of machine learning models. Researchers have used explainable machine learning methods to predict diseases in cattle have demonstrated that SHAP can be used to establish the most critical environmental and biological processes that drive disease risk [10].

This interpretability is especially helpful when it comes to veterinary decision-making, where domain experts demand an explanation of model predictions that are easy to comprehend to be able to plan disease control interventions.

Even with these developments, there are several limitations in the current livestock disease forecasting methods. Most current models assume comparatively dense and continuous surveillance data which is often unavailable in veterinary disease surveillance. Besides this, most of the models work on establishing disease occurrence or number of cases through single-stage frameworks of modelling which might not effectively explain the two underlying processes in epidemiological data; occurrence of outbreaks and then the severity of outbreak once it occurs. Moreover, despite growing integration of environmental factors into predictive models, comparatively limited research combines various environmental datasets at once or considers region-specific climatic regimes that mediate seasonal disease patterns.

The current research will solve these shortcomings by developing a climate-sensitive livestock disease forecasting framework that specifically considers sparse surveillance data. The suggested system re-creates a district-month dataset based on irregular epidemiological reports and incorporates multi-source environmental data, such as precipitation, temperature, humidity, vegetation indices, soil moisture, wind velocity, and the livestock density. Contrary to the conventional models of single stage prediction, the framework uses a two-stage machine learning architecture which is capable of prediction of probability of occurrence of an outbreak and secondly, prediction of the severity of the outbreaks in the event of an outbreak probability. Such modelling approach is also most appropriate in cases of zero-heavy disease data and enables the system to more effectively account for the complex interplay between the environmental conditions and disease transmission dynamics. Moreover, the system also uses explainable AI methods and uncertainty-awareness forecasting temporalities to generate interpretable and decision-supportive forecasts to veterinary officials.

## 1.3 Research Gap

Despite the extensive studies related to the prediction of infectious diseases and the detection of outbreaks, there are still several major drawbacks of livestock disease prediction systems. Currently, the literature covers the statistical detection algorithms of outbreaks, statistical models with zero-inflated and machine learning methods of forecasting outbreaks of infectious diseases. Nevertheless, most of these methods presuppose the availability of well-organized epidemiological data for similar temporal reporting.

In the veterinary disease surveillance systems, especially in the developing countries, such organized datasets are not very common. Veterinary epidemiology bulletins that are released by the Department of Animal Production and Health (DAPH) [11] are mostly used to monitor livestock diseases in Sri Lanka. Such reports usually include consolidated annual case numbers and months that the outbreaks took place at the district level as opposed to monthly consistent records of the cases. Therefore, the body of research on how sparse reports of surveillance can be converted into structured data to be used in machine learning-based forecasting is low.

Besides this, majority of the currently used models of prediction of livestock diseases are mainly based on past incidences of diseases and often do not incorporate various environmental factors that affect the spread of the diseases. The climatic and ecological conditions like rainfall, temperature, humidity, vegetation cover, soil moisture and livestock density may have a great influence on the survival of pathogens, the populations of vectors, and the movements of livestock. Nevertheless, even with this relationship, there are relatively few forecasting systems in which multi-source environmental datasets are used to model livestock disease risk holistically.

The other research gap that cannot be found in the existing body of research is the fact that modelling architectures to be used exclusively with zero-heavy epidemiological data has not been used extensively. The datasets of livestock diseases are full of intervals with no outbreaks and then the irregular outbreaks. In these datasets, common predictive models can be biased to predict zeroes and will not be able to record the rare but significant outbreak patterns.

The classical outbreak detection models include the Farrington model, the Early Aberration Reporting System (EARS), which are mainly based on statistical surveillance of past disease rates to identify abnormal increases in disease rates [1], [2]. Although such approaches are common in epidemiological surveillance, the methods presuppose high-density and uniform reporting systems. Comparative studies of outbreak detecting algorithms have revealed that their performance declines with irregular reporting and sparse observations in the dataset [3].

In dealing with the issue of overly zero observations researchers have suggested some statistical methods like zero-inflated and hurdle regression models to model the infectious disease data. These models explicitly factor out the occurrence of disease and disease intensity and have been shown to perform better on the large number of datasets with zero observations [4], [5]. As an example, a zero-infected negative binomial model has been implemented to enhance prediction of outbreak of the Foot-and-Mouth Disease in livestock populations [6]. Most of these strategies, however, continue to use historical records mainly on the number of diseases but fail to consider the environmental factors that contribute to the rise of diseases.

The prediction of disease in livestock is an activity that has recently been sought to be enhanced using meteorological variables through machine learning studies. Meteorological and geospatial characteristics have been predicted to have outbreaks of Lumpy Skin Disease, this has proven that the environmental conditions including rainfall, temperature, and vegetation indices were of great relevance to disease outbreaks [8], [9]. Random Forest and Gradient Boosting are ensemble learning approaches that have shown a high predictive performance in the modelling of the complex relationship between environmental variables and disease occurrence. However, most of these models have well-structured datasets as their assumptions and they do not explicitly deal with sparse surveillance data or the necessity to find interpretable prediction model.

Moreover, most systems based on machine learning are black-box based forecasting systems, which do not explicitly explain the factors affecting the likelihood of the disease occurring. Even though it is possible to apply explainable artificial intelligence methods like SHapley Additive exPlanations (SHAP) to enhance model interpretability in disease prediction [10],

these methods have not been utilized in integrating into climate-informed livestock disease forecasting systems.

To reduce the shortcomings that are presented by the literature, this paper presents a climate-aware seasonal livestock disease prediction framework, tailored to sparse veterinary surveillance data.

The first is that the proposed system proposes a data reconstruction mechanism that transforms infrequent epidemiological notifications into a structured district-month data that can be analyzed using machine learning. This allows us to do predictive modelling where the other monthly records of the cases are not known.

Second, the model incorporates multi-source environmental data, such as rainfall, temperature, humidity, vegetation indices, soil moisture, wind speed, and livestock density. The modelling approach informed by the climate enables the system to simulate environmental conditions that affect the action of disease transmission.

Third, the suggested architecture uses a two-step machine learning architecture that predicts the occurrence of an outbreak initially before estimating the severity of the outbreak in case an outbreak occurs. This modelling approach deals explicitly with the zero-heavy nature of epidemiological data and enhances the quality of representation of rare outbreak events.

Lastly, the system includes explainable artificial intelligence and uncertainty-aware prediction intervals, which can allow interpretable predictions and confidence estimates that can be used to assist veterinary decision-making.

The necessity to have better livestock disease forecasting systems can be evidenced by the organization of veterinary surveillance data in Sri Lanka. The most common method through which is the livestock disease monitoring conducted is the epidemiology bulletin, which is released by the Department of Animal Production and Health (DAPH) [11]. These reports give the case numbers per year consolidated with the months that an outbreak took place at the district level and not monthly case figures.

The consequence of this reporting structure is that there are sparse epidemiological data sets with irregular time-observation and high count of zero values. These features render challenging to use conventional statistical outbreak detection algorithms or many machine learning models that need regular time-series data. Thus, there is an evident requirement of prediction models that can recreate structured data based on sparse surveillance information and integrate environmental determinants of disease spread to enhance the prediction of outbreaks.

# 2. OBJECTIVE

## 2.1 Main Objective

The main objective of this study is to create a climate-sensitive livestock disease forecasting model to predict the likelihood and severity of an outbreak of the Foot-and-Mouth Disease (FMD) and Lumpy Skin Disease (LSD) in Sri Lanka, based on sparse veterinary surveillance records and environmental parameters.

The suggested system will reconstruct district-month epidemiological datasets from veterinary surveillance reports, and multi-source environmental data, including rainfall, temperature, humidity, vegetation indices, soil moisture, wind speed, and livestock density. A two-stage machine learning modelling approach will be implemented to estimate outbreak probability and outbreak severity while incorporating explainable artificial intelligence and uncertainty-aware prediction intervals to support veterinary decision-making.

This objective can be directly connected to the research gap that has been identified as a lack of livestock disease forecasting systems that can work with the sparse surveillance datasets and incorporate the environmental drivers of disease transmission.

## 2.2 Specific Objectives

To achieve the main objective, the following specific objectives are defined:

1. **To collect and preprocess epidemiological surveillance data for livestock diseases in Sri Lanka.**

This involves deriving district level reports of diseases in foot and mouth disease (FMD) and Lumpy Skin Disease (LSD) in Department of Animal Production and Health (DAPH) epidemiology bulletins and creating a structured data set of districts, year, outbreak months, cases, and deaths.

Table 3.2: Spatial and temporal distribution of FMD in 2022

| No | District | No of affected VS ranges | No of | | Months of occurrence |
|---|---|---|---|---|---|
| | | | Cases | Deaths | |
| 1 | Anuradhapura | 12 | 896 | 26 | January, April, May |
| 2 | Polonnaruwa | 1 | 176 | 2 | January |
| 3 | Kurunegala | 3 | 19 | 0 | January, February |
| 4 | Puttalam | 6 | 401 | 14 | January, February |
| 5 | Gampaha | 4 | 91 | 0 | January, February |
| 6 | Kalutara | 3 | 14 | 2 | January |
| 7 | Colombo | 2 | 143 | 51 | January |
| 8 | Mullativu | 2 | 85 | 2 | August, December |
| 9 | Mannar | 1 | 2 | 0 | October |
| 10 | Kandy | 4 | 80 | 10 | January, February, June |
| 11 | Matale | 1 | 32 | 13 | February |
| 12 | Ampara | 5 | 187 | 1 | March, August, December |
| 13 | Trincomalee | 6 | 716 | 23 | January, March, December |
| 14 | Batticaloa | 2 | 72 | 0 | March, April |
| 15 | Kegalle | 2 | 13 | 0 | February, May |
| 16 | Badulla | 3 | 154 | 18 | February, May, June, December |
| 17 | Monaragala | 1 | 50 | 2 | December |
| 18 | Hambantota | 1 | 35 | 0 | October |
| 19 | Galle | 1 | 3 | 0 | May |
| | All Island total | 60 | 3169 | 164 | |

*Figure 3: Example for DAPH FMD annual report data*

Table 3.3: Spatial and temporal distribution of LSD in 2023

| No | District | No of affected VS ranges | Number of | | Months of occurrence |
|---|---|---|---|---|---|
| | | | Cases | Deaths | |
| 1 | Anuradhapura | 8 | 874 | 0 | January , February, May, June, |
| 2 | Polonnaruwa | 3 | 270 | 10 | May, June, |
| 3 | Kurunegala | 21 | 2,731 | 27 | March, May, June, November |
| 4 | Puththalam | 7 | 768 | 3 | February,March,,Aprl, May |
| 5 | Gampaha | 3 | 20 | 0 | January, March,May |
| 6 | Kalutara | 3 | 59 | 0 | May June |
| 7 | Jaffna | 7 | 160 | 13 | January, Feb, March |
| 8 | Mullativu | 5 | 154 | 2 | February, March |
| 9 | Mannar | 2 | 64 | 0 | March |
| 10 | Vavuniya | 4 | 511 | 3 | March |
| 11 | Kilinochchi | 2 | 465 | 48 | January, March |
| 12 | Kandy | 5 | 74 | 2 | February ,June, July |
| 13 | Mathale | 1 | 122 | 2 | April |
| 14 | Nuwara Eliya | 4 | 189 | 15 | October, November, December |
| 15 | Ampara | 11 | 769 | 23 | March, May , June, July |
| 16 | Batticaloa | 09 | 889 | 15 | March, May, June, October |
| 17 | Trincomalee | 08 | 343 | 17 | May |
| 18 | Kegalle | 03 | 33 | 2 | May, June |
| 19 | Rathnapura | 6 | 149 | 0 | May, June, September, October |
| 20 | Badulla | 11 | 506 | 26 | May, June, July, August, October |
| 21 | Monaragala | 10 | 1017 | 8 | February, May, June, July |
| 22 | Hambantota | 2 | 5 | 0 | September, October |
| 23 | Matara | 3 | 23 | 0 | September, October |
| 24 | Galle | 3 | 100 | 0 | June, August |
| | Total | 141 | 10,295 | 216 | |

*Figure 4: Example for LSD DAPH annual report data*

2. **To collect and integrate multi-source environmental and climatic datasets relevant to livestock disease transmission.**

Separate datasets on the environment (CHIRPS rainfall, NASA POWER meteorological variables (temperature and humidity), SMAP soil moisture, MODIS vegetation index (NDVI), wind speed data, and livestock density data) will be gathered and combined with the epidemiological dataset.

*Table 1:Data Sources and Integrated Variables*

| Variable Category | Specific Dataset / Source | Data Link |
|---|---|---|
| Rainfall | **CHIRPS** (Climate Hazards Group InfraRed Precipitation with Station data) | https://www.chc.ucsb.edu/data/chirps |
| Meteorological (Temperature/ Humidity/ Wind speed) | **NASA POWER** (Prediction of Worldwide Energy Resources) | https://power.larc.nasa.gov/data-access-viewer/ |
| Soil Moisture | **NASA SMAP** (Soil Moisture Active Passive) | https://nsidc.org/data/smap |
| Vegetation Index (NDVI) | **MODIS** (Moderate Resolution Imaging Spectroradiometer | https://www.earthdata.nasa.gov/data/catalog/lpcloud-mod13a2-061 |
| Livestock Density | **FAO GLW** (Gridded Livestock of the World) | https://www.fao.org/livestock-systems/global-distributions/en/ |

3. **To reconstruct a district–month machine-learning-ready dataset from sparse surveillance reports.**

The study will make aggregated disease reports into a structured time series data that will reflect the occurrence of outbreaks and environmental conditions in each district and month.

4. **To develop a two-stage machine learning modelling framework for livestock disease forecasting**.

The former will provide predictions of the likelihood of disease outbreaks occurrence, whereas the latter will involve estimating the severity of the outbreak with the help of appropriate machine learning classification models.

5. **To incorporate explainable artificial intelligence techniques into the forecasting framework.**

The experience of SHAP-based model explainability will be applied to help define the environmental and epidemiological factors that contribute to the risk of prediction of disease and the severity of an outbreak.

6. **To quantify prediction uncertainty in livestock disease forecasts.**

Prediction intervals based on bootstraps will be created to produce estimates of uncertainty in the prediction of the severity of the outbreak and give confidence interval to decide.

7. **To evaluate the performance of the proposed forecasting framework using time-aware validation methods.**

The developed models will be evaluated on forward-chaining validation, and the performance metrics include F1-score, precision, recall, PR-AUC, and classification accuracy to determine the effectiveness of the developed models in predicting livestock disease outbreaks.

# 3. METHODOLOGY

## 3.1 Research Approach

The study is based on Design Science Research (DSR) framework of developing and testing a climate-informed forecasting system of livestock diseases. Design science will be suitable since its main goal of the project is to design, implement, and test a predictive system that can accurately predict the outbreak of livestock diseases using epidemiological and environmental information.

The proposed system is expected to overcome the shortcomings of the current model of predicting livestock diseases, especially the failure to efficiently use a sparse surveillance dataset and environmental drivers of disease spread. The study is thus aimed at developing a new forecasting framework that is built around epidemiological data reconstruction, environmental data fusion, machine learning modelling, explainability, and uncertainty estimation.

The system will be modeled as a two-step machine learning prediction system that forecasts the occurrence of the outbreaks and the severity of the outbreaks independently. The reason why this method is chosen is to deal with the zero-heavy structure that is often present in epidemiological data, with many observations having no outbreaks.

## 3.2 Technical Methods and Algorithms

### 3.2.1 Dataset Construction

The initial methodology approach is the creation of a district-month epidemiological dataset by analyzing veterinary surveillance reports. Data on livestock diseases will be obtained using data on Department of Animal Production and Health (DAPH) epidemiology bulletins which give information on outbreaks at the district level including the total case, months of outbreak, and number of deaths.

Since these reports do not contain monthly records on cases, but only provide aggregated outbreak data, the data will be converted into a structured dataset, with each row containing a district-month observation. The presence or absence of an outbreak will be coded as a binary variable that will tell whether a disease outbreak was experienced that month or not.

The environmental and climatic variables will then be added to the dataset. The research work will gather environmental data using various sources which are publicly available, such as:

- CHIRPS dataset for rainfall measurements
- NASA POWER dataset for temperature, humidity, and wind speed variables
- SMAP satellite dataset for soil moisture
- MODIS satellite data for vegetation indices (NDVI)
- FAO Gridded Livestock of the World dataset for cattle density.

These variables are chosen since past research has indicated that environmental factors have an overpowering impact on the dynamics of livestock diseases, especially the diseases that are transmitted through vectors such as Lumpy Skin Disease.

### 3.2.2 Feature Engineering

To fit environmental patterns that drive disease transmission, several feature engineering methods will be used.

To begin with, the inclusion of environmental variables will be done as monthly climate characteristics such as rainfall, temperature, relative humidity, vegetation index, soil moisture, wind speed, and livestock density.

Second, lag features will also be produced to capture a delayed effect of the environment on the transmission of the disease. Let us take the example of rainfall and vegetation patterns of past months which can impact on the coming months about populations of the vectors and risk of diseases.

Third, the use of seasonal encoding will be used to capture the cyclical seasonal trends. Month values will be mathematically transformed into sine and cosine values to capture the seasonal trends without setting any discontinuities at the end of December and January.

Also, aspects of the seasons associated with monsoons will be incorporated to reflect the Southwest monsoon, North-east monsoon and inter-monsoon seasons that have a significant impact on the climate pattern in Sri Lanka.

### 3.2.3 Two-Stage Machine Learning Forecasting Framework

The proposed forecasting system will employ a **two-stage modelling architecture** to address the zero-heavy structure of livestock disease datasets.

**Stage 1: Outbreak Occurrence Prediction.**

The former stage predicts the likelihood of occurrence of disease outbreak in a district and month. This step considers the issue as a dichotomous classification problem.

The following are candidate algorithms:

- Logistic Regression (control model)
- Random Forest classifier
- The Gradient Boosting classifier.
- XGBoost classifier

The reason why random forest and boosting-based algorithms are chosen is that both are suitable in structured tabular data as well as nonlinear feature interactions. It has been established in the past that using ensemble tree-based models is effective in the disease prediction task that involves the environmental variables.

**Stage 2: Outbreak Severity Prediction**

In case an outbreak is forecasted during Stage 1, the second stage approximates the level of severity of the outbreak. The severity will be categorized into Low, Medium, or High according to a severity score expressed in the number of cases, the period of an outbreak, and mortality rates.

Random Forest and XGBoost classifiers will be evaluated for this stage due to their ability to handle multiclass classification tasks and nonlinear relationships between environmental features and disease severity. This two-stage modelling framework enhances the accuracy in prediction in that a distinction is made between prediction of the occurrence and intensity of the outbreak, which in most cases is controlled by various factors.

## 3.3 Tools and Platforms

Python will be used to implement the forecasting framework because of its wide variety of data science, machine learning, and geospatial analysis.

The most important libraries and tools are:

- **Python (primary programming language)**

Python is chosen because it has good support of data science processes, machine learning libraries and is used in scientific studies by a large percentage.

- **Pandas and NumPy**

Applied in preprocessing of data, building of datasets and feature engineering.

- **Scikit-learn**

Offers applications of machine learning models like the Logistic Regression and Random Forest, and model evaluation tools.

- **XGBoost**

Selected since gradient boosting algorithms can be very high performing on structured tabular data and can model a complex nonlinear association.

- **SHAP (SHapley Additive exPlanations)**

Utilized in model interpretability. SHAP is chosen due to its attribute consistency and ability to enable the researcher to determine the most contributing factors in the environment to predict diseases.

- **Bootstrap Prediction Intervals**

Bootstrap resampling will be used to estimate prediction uncertainty. This approach is chosen because it provides a simple and model-agnostic method for generating prediction intervals without requiring distributional assumptions.

## 3.4 Data Collection and Ethical Considerations.

The data sets employed in this study will be taken out of the open sources. The epidemiological information will be gathered using the reports of the Department of Animal Production and Health (DAPH) that are publicly available in government publications.

Global climate and satellite repositories (CHIRPS, NASA POWER, MODIS, and SMAP) will be used to gather additional environmental data. These data are publicly available and popular among environmental and epidemiological studies.

As far as the study is based solely on secondary data that is publicly available and does not involve any human participants or personally identifiable data, the study does not create significant ethical issues regarding privacy or informed consent. Yet all data sources will be given proper attribution.

## 3.5 Strategy and Performance Metrics of validation.

A time-conscious validation strategy will be used to assess model performance to prevent leakage of data as well as realistic performance of forecasting.

The study will not require random dataset splitting but will use forward-chaining validation where historical data will be used as the model train and future periods as the test data. This is more realistic as far as real-life forecasting is concerned.

The following measures will be used to determine performance:

- Precision
- Recall
- F1-score
- ROC-AUC
- PR-AUC

Recall and PR-AUC are especially relevant to predicting the outbreak because a missed true outbreak (false negative) can have severe implications to the control of livestock diseases.

To predict severity, we will use other measures like confusion matrices and macro F1-scores to evaluate the performance of classification in the different categories of severity.

Another criterion that will be used to determine the final model is predictive performance, stability, interpretability and the ability to work with sparse epidemiological data.

# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

## 4.1 Overview of the Proposed Solution and Its Components

The Animal Disease Reporting System (ADRS) is a centralized, multi-modal intelligence platform designed to transition livestock management from a reactive to a proactive paradigm. The system is composed of four primary research-driven components: Seasonal Disease Forecasting (DAPH disease reports and multi-source environmental data), Real-Time Geospatial Visualization (mapping spread velocity), Wellness & Management Standardization (health grading and BCS), and the Computer Vision Diagnostic & Prognostic Assessment module.

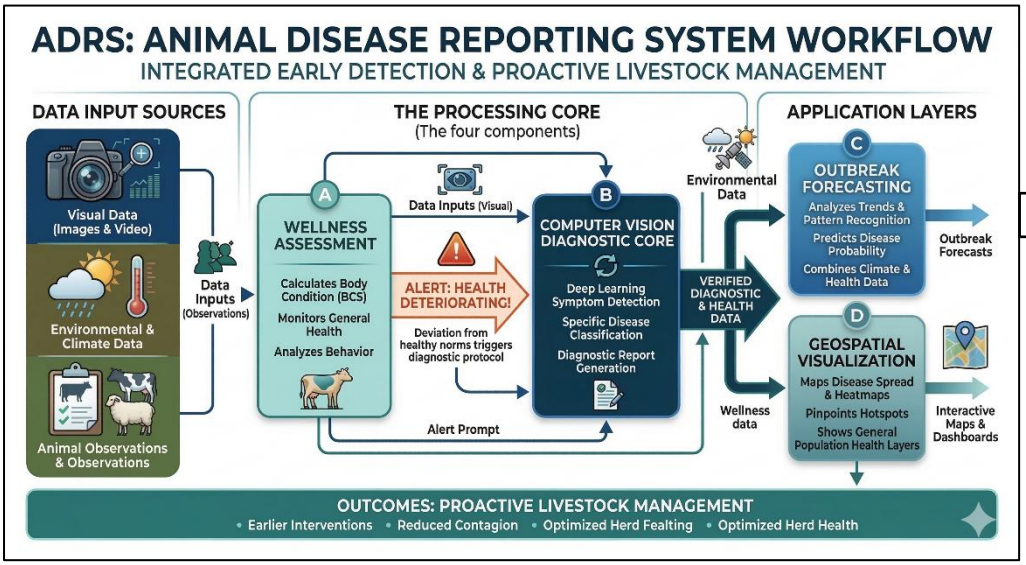



*Figure 5: System Diagram with Components*

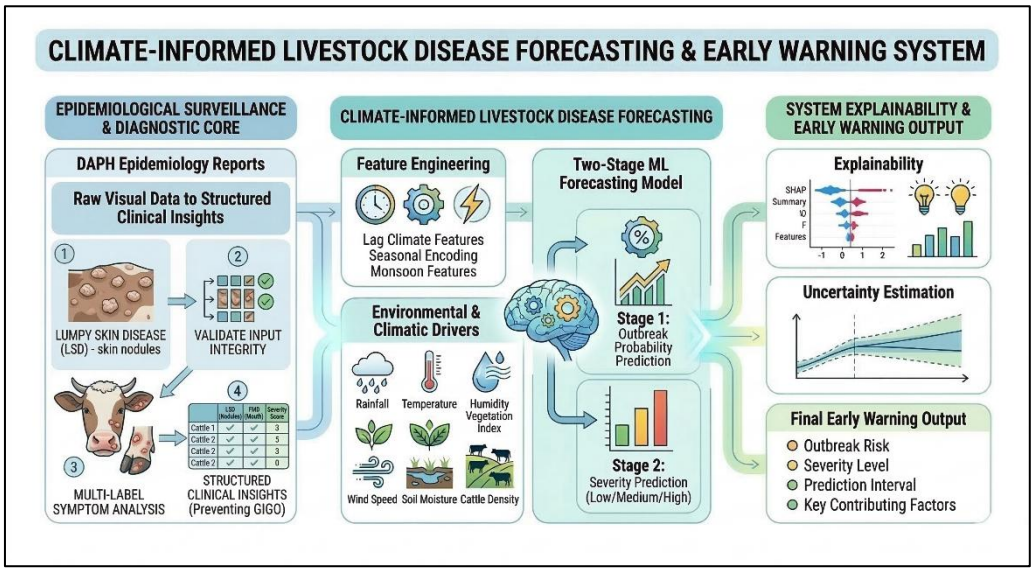



*Figure 6: How Disease forecasting and early warning system fits into the system*

This research is based on one of these four components which is the Seasonal Disease Forecasting and Early Warning component. This part will convert sparse DAPH disease and multi-source environmental data to predictive disease outbreak intelligence by reconstructing data, feature engineering, two-stage machine learning, uncertainty estimation, and explainable output generation. The system direction is also in line with the project plan, in which the disease reports and climate data will first be combined, and then be utilized to make predictions on outbreaks, severity, uncertainty, and generation of explanations.

## 4.2 Highlight of the Individual Component and Its Parts

The individual research part is the Seasonal Disease Forecasting and Early Warning component. It is primarily used to predict any outbreak of livestock diseases in the future like FMD and LSD based on sparse surveillance data and environmental variables. This component, as compared to the previous ones, which have been primarily involved in the present-condition evaluation and the current-case intelligence, provides the future prediction of an outbreak.

This component contains six main sub-parts:

**1. Dataset Construction Module**

Data sources that are used in this module to gather epidemiological indicators are DAPH bulletins, and the environmental indicators are CHIRPS, NASA POWER, SMAP, MODIS NDVI, and livestock density datasets. It rebuilds these inputs into machine-learning-ready data of district-month.

**2. Feature Engineering Module**

This module derives predictive inputs based on the raw input which are climate variables, lag features, cyclical seasonal encodings and monsoon-informed features. These characteristics are supposed to embrace both the immediate and delayed environmental impacts of disease propagation.

**3. Stage 1 Outbreak Prediction Module**

This module foresees the occurrence or not of an outbreak in each district and month. It approaches the problem as a binary classification problem.

**4. Stage 2 Severity Prediction Module**

In the case of an outbreak that is predicted in Stage 1, this module approximates the severity of an outbreak, usually Low, Medium or High. The reason why this two-stage methodology is employed is that the surveillance data are sparse and zero-heavy and it is less reliable to directly predict the cases using just the single stage.

**5. Uncertainty Estimation Module**

This module uses bootstrap prediction intervals to estimate the confidence range of forecast outputs. This improves the usefulness of the model for decision-making, because veterinary officers can see not only the forecast but also the uncertainty associated with it.

**6. Explainability Module**

In this module, SHAP-based explainability is used to determine the most significant environmental and epidemiological characteristics that contribute to every prediction. This enhances transparency and increases the forecasting output to be more interpretable by the veterinary authorities.

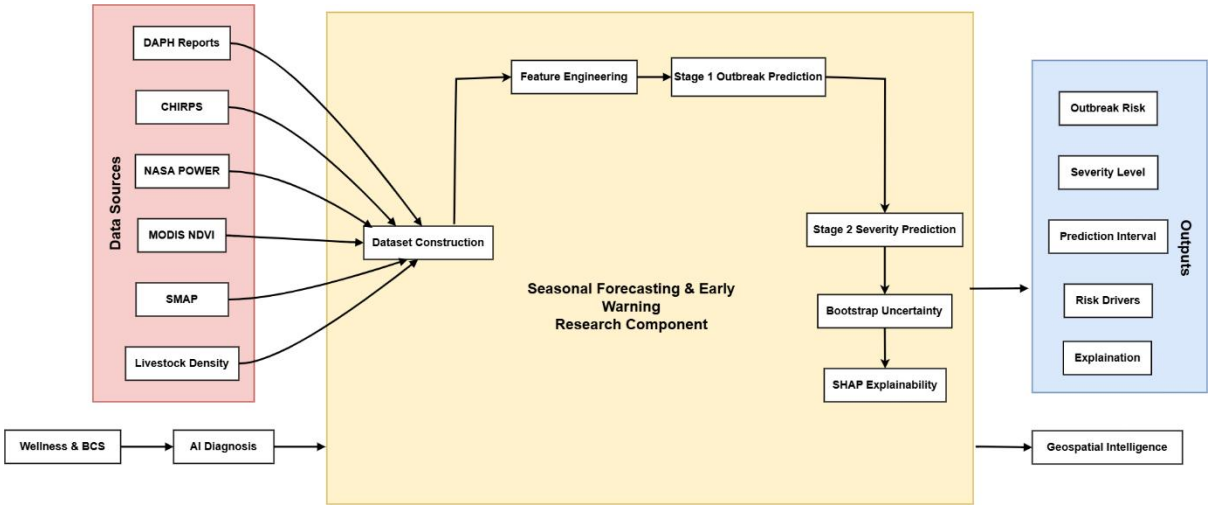



*Figure 7: system architecture diagram for seasonal forecasting component*

## 4.3 Component Interactions

The four system elements engage a logical surveillance-to-forecasting chain. The Wellness and BCS component are a component that captures primary health indicators and assists in early intake. The AI Smart Diagnosis component aids in the detection of probable disease cases based on the observed symptoms or picture. The Geospatial Intelligence component then locates cases that are detected or reported within a spatial environment to aid in the plan cluster detection and containment. Finally, the Seasonal Forecasting component uses accumulated surveillance information together with environmental and climatic data to generate district-level early warning forecasts. In this way, the forecasting component extends the system from current surveillance to future risk prediction.

Within the forecasting component itself, interaction occurs as a pipeline: the dataset construction module sending its output to feature engineering; engineered features sent to Stage 1 outbreak prediction; positive outbreak prediction sent to Stage 2 severity prediction; and uncertainty estimation and explainability applied to the final forecasts before producing decision-ready output.

## 4.4 Data Flow Explanation

The proposed architecture has its data flow starting with gathering of two large data types of namely epidemiological surveillance and environmental data. Different data sets that are used to acquire epidemiological data include DAPH veterinary bulletins and external climate and satellite data sets which are used to acquire environmental variables. These crude inputs are

initially purged and matched during the dataset building phase to generate a structured district-month dataset.

This combined dataset then gets converted by feature engineering into modelling features which include rainfall lags, humidity lags, NDVI, soil moisture, monsoon phase, and seasonal encodings. These characteristics are further transferred to Stage 1 model which forecasts the occurrence of an outbreak. The Stage 2 model uses the same feature set in case severity needs to be predicted in case of an outbreak. Following the prediction step, bootstrap-based prediction intervals are computed by the uncertainty estimation module, and the most influential factors are identified by the explainability module based on SHAP. Lastly, the system generates an early warning that is comprised of probability of outbreak, degree of severity, uncertainty range, and the variables of paramount importance.

## 4.5 Scalability and Security Considerations.

The proposed system can be extended, i.e., it is scalable since it is possible to add new diseases, districts, or environmental variables without altering the entire architecture. Future work can also expand the framework to larger windows or larger area coverage by the district-month modelling structure. In the same manner, the data layer can be updated with new climate or livestock-related data as soon as they are available.

In terms of security, it is primarily based on aggregated veterinary surveillance data and open environmental data, and hence the threats to privacy are minimal. Nevertheless, sensitive information management procedures must be followed even in storing the records of DAPH derivations and the forecasts made. Only authorized veterinary or administrative personnel should get access to system outputs when they are to be used in a future decision-support dashboard. Moreover, all the external datasets and government reports incorporated into the framework should be appropriately attributed.

# 5. USER REQUIREMENTS

## 5.1 Stakeholder Identification

The suggested Climate-Informed Livestock Disease Forecasting and Early Warning System has a variety of stakeholders who engage with the system in various ways. These stakeholders consist of direct users, as well as organizations which receive the fruit of the forecasting outputs.

1. Veterinary Officers (Primary Users)

The major users of the system are government veterinary officers who monitor livestock diseases at the district and regional level. With the help of the risk prediction of the outbreaks and the severity estimates, they plan the preventive measures, field inspection, and efficient allocation of veterinary resources.

2. Department of Animal Production and Health (DAPH).

National livestock disease surveillance and epidemiological reporting are done by DAPH. The suggested forecasting element can assist DAPH with the early warning of possible outbreaks on environmental and climatic drivers.

3. Livestock Farmers

Indirectly, the farmers are helped by the system because of the early warning and prevention advice provided by veterinary authorities. The farmers can be assisted by early forecasting to take preventive measures, e.g. vaccination or movement control.

4. Agricultural Policy Makers and Decision Makers.

The forecasting outputs can be used by government authorities and policy makers to plan the disease control strategies, allocate veterinary budget, and institute regional disease management programs.

5. Scientists and Statistical Analysts.

The outputs and models of the system can be further studied and used by researchers in the fields of veterinary epidemiology and agricultural data science to analyze and improve the predictive models and conduct future studies on the risk of disease occurrence.

## 5.2 Functional Requirements

The functional requirements explain the basic functions that the system should offer to aid in forecasting livestock diseases and early warnings.

**1. Data Collection from Epidemiological Reports.**

The system will obtain livestock disease surveillance information in DAPH epidemiology bulletin, such as quarter, year, outbreak months, number of cases and number of deaths.

**2. Integration of Environmental Datasets.**

The system will combine the environmental and climatic data like rainfall, temperature, humidity, vegetation index (NDVI), soil moisture, and density of livestock with the epidemiological one.

**3. Dataset Reconstruction for Machine Learning.**

The system will rebuild the thinly spread epidemiological data into a machine learning-friendly district-month dataset.

**4. Feature Engineering for Disease Prediction**

The system will produce predictive variables such as lag climate variables, seasonal encodings, and monsoon-based indicators to represent environmental factors on disease outbreaks.

**5. Outbreak Prediction**

The system will estimate the likelihood of a livestock disease outbreak in every district and month with the aid of machine learning.

**6. Outbreak Severity Prediction**

The system shall be able to estimate severity of predicted outbreaks based on severity categories like Low, Medium and High.

**7. Uncertainty Estimation**

The system will produce prediction ranges by incorporating bootstrap techniques to show the uncertainty of predictions in the models.

**8. Explainable Predictions**

The system will be able to give explainable outputs through SHAP to establish the influential environmental factors of predicting an outbreak.

**9. Risk Output Generation**

The system will be able to produce outputs that are ready to be given such as the probability of an outbreak, level of severity, predicted interval and the major risk factors.

## 5.3 Non-Functional Requirements

Non-functional requirements detail the quality features that the system should meet to become reliable and usable.

1. **Performance Requirements**

The system must also be able to handle environmental and epidemiological data in an efficient manner and provide forecasts of outbreaks in a reasonable time frame that can be used in the decision-making process.

2. **Scalability Requirements**

Design The design must be able to accommodate the addition of new datasets, diseases or geographic regions without a significant architecture redesign.

3. **Security Requirements**

The system should provide safe management and storage of epidemiology data received through veterinary authorities. The process of assessing the forecasting output should be restricted to authorized users.

4. **Usability Requirements**

The system must show the outcomes of prediction in easy and comprehensible format to ensure that the risk of outbreak is interpreted easily by veterinary officers and decision makers.

5. **Reliability Requirements**

The system must yield the same predictions when a repetition of the system with the same input data and modelling framework are used.

6. **Maintainability Requirements**

The system architecture should allow models and datasets to be updated as new surveillance data becomes available.

# 6. COMMERCIALIZATION PLAN

The given Climate-Informed Livestock Disease Forecasting and Early Warning System can be commercialized as a decision-support tool to be used by veterinary officials and livestock industry actors. The system is used to obtain predictive intelligence on the outbreak of livestock diseases by integrating epidemiological surveillance data with climatic and environmental conditions. The system can assist the veterinary agencies to change their reactive response to outbreak containment to proactive prevention of disease and its resources planning by converting raw surveillance data into practical early-warning information.

The commercialization approach focuses on institutional customers such as government veterinary departments, livestock organizations, and commercial dairy operations that require reliable livestock disease risk forecasting tools.

## 6.1 Target Market and Customer Persona

The forecasting system has a primary market in Business-to-Government (B2G) market, especially the government institutions mandated with surveillance and control of livestock diseases.

**Primary Market**

The primary target customer is the Sri Lankan Department of Animal Production and Health (DAPH) and the provincial veterinary divisions which run livestock disease control programs. Such organizations need quality forecasting tools to monitor the trend of diseases and ensure that they effectively allocate veterinary resources.

Thousands of dairy farms and cattle owners are spread throughout all districts in the livestock sector in Sri Lanka and this has given rise to the high demand of enhanced disease surveillance and prediction systems.

**Secondary Market**

The secondary market is made up of Business-to-Business (B2B) customers including:

- dairy cooperatives
- commercial livestock farms
- private veterinary clinics
- livestock research institutions

Such organizations seek predictive answers to control herd health and prevent losses of diseases and enhance productivity.

**Customer Persona**

A representative customer persona for this system is a senior veterinary epidemiologist working in a government livestock surveillance unit, such as a veterinary officer or veterinary academic involved in outbreak monitoring and disease risk assessment.

An epidemiologist of the veterinary, such as Dr. Thusitha K. Karunarathna (University of Peradeniya), specialized in the epidemiology of infectious diseases and surveillance, would be interested in such system, which would allow them to get reliable predictions of an outbreak and clarifications of environmental risks. Practitioners in this position may experience difficulties because of the lack of surveillance information, slow reporting of outbreaks and absence of predictive analytics concerning the early identification of the disease. The suggested forecasting system will resolve these issues by incorporating climatic data with machine learning to produce proactive outbreak risks and interpretable insights to assist in decision making.

## 6.2 Value Proposition

The value proposition of the proposed forecasting component can be summarized as:

**“Climate-informed early warning for livestock disease outbreaks.”**

The system offers some of the major advantages:

1. Proactive disease management.

Rather than responding when the outbreaks have already happened, the veterinary authorities can predict the high-risk times and take preventive measures to avoid the problem of outbreaks, including vaccination or movement restrictions.

2. Data-driven decision support.

The system merges epidemiological and environmental data, thereby identifying environmental factors that predispose people to outbreaks.

3. Efficient resource allocation.

Surveillance and intervention at higher risk districts can be given priority by the veterinary authorities.

4. Explainable predictions

SHAP explainability allows the veterinary officers to gain knowledge on why a particular district is likely to suffer higher risk of an outbreak.

5. Uncertainty-aware forecasting

Bootstrap prediction intervals give an estimate of confidence of the forecast that can enhance dependability of the decision making.

The system assists in the more efficient livestock disease control and enhanced agricultural production by converting sparse surveillance data into actionable predictions.

## 6.3 Revenue Model

Commercialization strategy employs hybrid models of business that integrate enterprise licensing and subscription-based services.

**Government Sector (B2G)**

The forecasting system would be accessed by government veterinary agencies including DAPH by an annual enterprise license agreement. Such an agreement would enable the authorized veterinary officers to have access to the forecasting platform and get predictions of the outbreaks of all the districts.

- The enterprise license would include:
- access to the forecasting dashboard
- periodic model updates and retraining

**Private Sector (B2B)**

The system is available to private organizations like dairy cooperatives and commercial farms on the Software-as-a-Service (SaaS) subscription.

The subscription tiers could be based on:

- number of farms monitored
- number of disease prediction queries per month
- level of analytical reporting

The livestock industry can adopt this model on a scale as the industry grows.

## 6.4 Pricing Strategy

Pricing policy is based on the value-based pricing and penetration pricing.

In the case of the government sector, an annual enterprise license that is subsidized will be provided to adopt it in national veterinary programs.

In the case of private organizations, there will be the implementation of a level subscription with dependability on the size of the farms as well as the level of use.

Example pricing structure:

*Table 2: Example pricing structure*

| Package | Monthly cost | Target Users |
|---|---|---|
| Basic | Rs. 2000 | Small farms |
| Standard | Rs. 5000 | Dairy cooperatives |
| Enterprice | Rs. 15000 | Large livestock companies |

forecasting solution is very economical, as this price is much lower than the price of traditional methods of monitoring precision livestock.

## 6.5 Competitive Advantage

The proposed forecasting system offers several advantages compared with existing livestock monitoring approaches.

*Table 3: Comparative Advantages of the Proposed Climate-Informed Forecasting Framework*

| Feature | Proposed Forecasting System | Traditional Reporting | Generic Data Analysis tools |
|---|---|---|---|
| Predictive outbreak forecasting | Yes | No | Limited |
| Climate-informed modelling | Yes | No | Rare |
| Uncertainty estimation | Yes | No | Rare |
| Explainable predictions | Yes | No | Limited |
| Low infrastructure cost | Yes | Yes | Moderate |

The Traditional veterinary surveillance system is mainly based on retrospective reporting and manual analysis. The suggested system will come with predictive machine learning models that incorporate environmental drivers, which would help in early detection of high-risk outbreak conditions. This predictive modelling coupled with explainability, and uncertainty estimation has a very competitive advantage over existing livestock surveillance approaches.

## 6.6 Intellectual Property Considerations

The proposed forecasting system has intellectual property such as the machine learning modelling framework, data processing pipeline, and feature engineering methods applied to climate-informed disease prediction.

Key IP elements include:

- the two-stage outbreak forecasting architecture
- the dataset reconstruction method for sparse surveillance data
- the integration of climate features and monsoon-based variables
- the uncertainty estimation and explainability modules

These elements are defensible in the form of software intellectual property and algorithm proprietary. Besides this, the curated data set that is created through the combination of both epidemiological and environmental data can be a valuable research resource.

Controlled access, securing data storage and confidentiality agreements are part of data governance practices that are to be adopted to avoid unauthorized use of proprietary models and datasets.

## 6.7 Cost Estimation

The cost of developing and operating the forecasting system can be divided into two categories:

1. **Development and model training costs**
2. **Operational deployment costs**

*Table 4: Cost Estimation*

| **Component** | **Estimated Cost** |
|---|---|
| Data collection and preprocessing | Rs. 25,000 |
| Cloud computing for model training | Rs. 4,000 / month |
| API deployment and server hosting | Rs. 6,000 / month |
| Data storage and backup | Rs. 2,500 / month |
| Internet / Data Packages for Data Collection | Rs. 3,000 / month |
| AI Tools / Software Subscriptions (e.g., ChatGPT, coding assistants) | Rs. 7000/month |
| Research & Development Labor | Rs. 160000/8 months |

# 7. BUDGET AND JUSTIFICATION

The implementation and testing of the suggested Climate-Informed Livestock Disease Forecasting System will need the resources connected with computing infrastructure, software applications, cloud computing, and human resources. Given the fact that the proposed system is mainly a software-based predictive analytics platform, the hardware needs are low as opposed to the traditional livestock monitoring systems, which use costly sensors or special devices. The budget is approximated to include the resources needed to develop and train a machine learning model and deploy it to a cloud and maintain its operation.

*Table 5: Hardware Cost*

**Hardware Costs**

| Item | Estimation |
|---|---|
| Development Laptop / Workstation (existing infrastructure) | Rs. 0 |
| External Storage / Backup Devices | Rs. 10,000 |

Justification: The suggested system will not need special hardware equipment in the form of sensors, IoT collars, and imaging devices. The development may be done with the help of an available workstation or a laptop. Nevertheless, to store environmental datasets, model artifacts and backup copies of processed datasets external storage might be needed.

**Software Licenses**

*Table 6: Software Licenses cost*

| Software / Tool | Estimated Cost |
|---|---|
| Programming Tools (Python, VS Code, Jupyter) | Free |
| Data Processing Libraries (Pandas, Scikit-learn, XGBoost) | Free |
| AI Development Tools (ChatGPT / AI assistants) | Rs. 7,000/month |

Justification: Most of the machine learning and data science tools applied in this study are open-source and free of charge. Libraries like Scikit-learn, Pandas and XGBoost in Python offer strong machine learning functionalities without incurring a fee. Non-essential AI development systems like ChatGPT or coding assistants, however, can come at a subscription fee to enhance the development productivity and to facilitate experiments with models.

**Cloud Services**

*Table 7: Cloud Services cost*

| Service | Estimated cost |
|---|---|
| Cloud Training of Models | Rs. 6000/month |
| Server Hosting and API Deployment | Rs. 6000/month |
| Cloud Storage and Backup | Rs. 3000/month |
| Internet Data Packages | Rs. 3000/month |

Justification: Cloud infrastructure is necessary to enable scalable model training, dataset processing and prediction services deployment. This is made possible by cloud computing resources which enable the system to compute environmental datasets and train machine learning models. Hosting services make it possible to deploy the forecasting model as API or web-based decision support system. Cloud storage is used to store and backup datasets and trained model artifacts based on security.

**Human Effort Cost**

*Table 8: Human Effort Cost*

| Activity | Estimated cost |
|---|---|
| Data Collection and Dataset Preparation | Rs. 25,000 |
| Development of the machine learning model | Rs. 160,000 |
| System Integration and Deployment | Rs. 100,000 |

Justification: The largest cost of the system developing process is human labor. The development activities encompass gathering epidemiological and environmental data, preprocessing data, building machine learning models, executing the feature engineering

pipelines, assessing the model, and deploying the prediction services. Software engineers and data scientists are needed in these activities.

Though the research is done as a part of an academic project, these costs reflect the approximate estimate of the value of labor that will be needed to develop and maintain the forecasting system in a practical implementation case.

# 8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) for the Animal Disease Reporting System (ADRS) was created to provide a way to distribute work among the research team and ensure that they are developing each component of the ADRS at the same time so that everything has the same timeline. The project is broken out into 4 components of research assigned to 1 researcher per component. The research and development of the ADRS will take place over a single academic year or 9 months, from the initial data collection to the complete integration of the system.

## 8.1 Task breakdown by member

**Component 1: AI-Powered Smart Diagnosis (Computer Vision and Prognostics)**

- Member 1: A.L.M Athulathmudali
- Key Tasks:

1. Collect and Annotate Primary and Secondary Data Sets for Local Cattle Breeds (i.e. Cattle Breeds found locally) for multiple concurrent diseases and their associated vitality indicators.
2. Design and Build Input Integrity Gate using YOLOv10-tiny and Out-of-Distribution (OOD) Additional Information.
3. Train Hierarchical Multi-Label Classifier (Swin Transformer) and Severity Quantification Module (Mask R-CNN).
4. Implement the Chain of Thought (CoT) transparent framework and prognostic survival engine.
5. Validate the diagnostic threshold using K-Fold cross-validation to ensure it has an 85% accuracy rate.

**Component 2: Real-Time Geospatial Clustering & Trajectory Mapping**

- Member 2: G.A Sandaru
- Key Tasks:

1. Gather and process both geographical and historical outbreak data regarding Sri Lanka's livestock.

2. Create density-based spatial clustering algorithms to detect contemporary (real-time) disease hot-spots.
3. Utilize physics-based spread modeling to determine both the localized outbreak velocities and trajectories.
4. Build an Interactive Intelligence Map for use by Government and Veterinary Agencies.
5. Build API endpoints to securely accept diagnostic payloads (type of disease and its severity) from the Computer Vision component.

**Component 3: Proactive climate-informed livestock disease forecasting system**

- Member 2: S.S Kumarasinghe
- Key Tasks:

1. Gather and processed epidemiological data based on the DAPH veterinary bulletins, incorporating the district level outbreaks, cases, deaths and months of outbreak of diseases like FMD and LSD.

2. Obtain and incorporate datasets in atmospheric environmental factors of livestock disease transmission, such as rainfall (CHIRPS), climate (NASA POWER), vegetation index (NDVI), soil moisture (SMAP), and livestock density.

3. District month machine learning data Construct a district-month machine learning data by transforming sparse surveillance reports into a structured temporal data set to predict modelling.

4. Carry out feature engineering of the climate-informed forecasting by creating lag climate features, seasonal encodings, and features of monsoons and the environment.

5. Build a sequential machine learning forecasting system to both predict the occurrence of outbreaks and estimate the magnitude of the outbreaks utilizing models like the Random Forest or XGBoost.

6. Apply uncertainty estimation and explainable AI techniques with bootstrap prediction intervals and SHAP feature attribution.

7. Measures Performance of forecasting models with forward-chaining validation and precision, recall, F1-score, and ROC-AUC.

**Component 4: Animal Health Detection (Wellness and Management)**

- Member 4: W.M.P.J Wijenayake
- Key Tasks:

1. Develop and establish corporate standard Multiple Base Parameters for the recognition of physiological distress and physical wellness in local dairy cattle.
2. Create a 2D vision-based algorithm to automate the reading of Body Condition Scores (BCS) by using skeletal key point detection software.
3. Use the clean segmented masks of bovine animals created using algorithms from the Computer Vision Component to remove lesions when calculating the BCS.
4. Create a farmer view of the wellness of the animal that includes recording the physiological conditions and energy status of the herd.
5. Conduct field tests to compare automated reading of BCS to manually recorded readings prepared by veterinarians.

## 8.2 Timeline and Workload Distribution

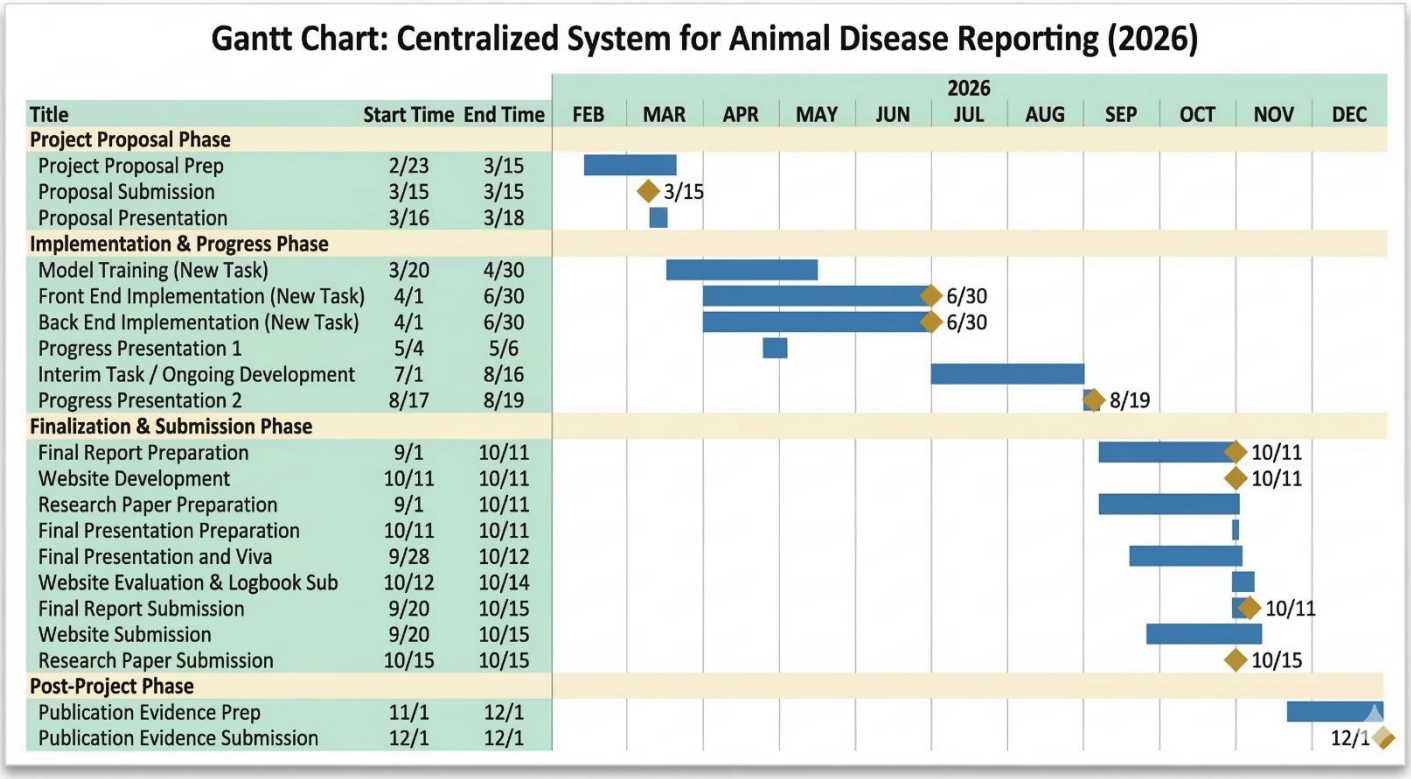
To ensure the workload is distributed evenly, each of the 4 Research Team members will complete the core software engineering phases simultaneously. The concurrent development methodology will help avoid any integration bottlenecks, as well as ensure that all module dependencies are available before the scheduled due date.

*Table 9: Timeline Alignment and Phase Distribution*

| **Project Phase** | **Timeline** | **Primary Activities (All Members)** | **Milestones / Deliverables** |
|---|---|---|---|
| **Phase 1: Requirements & Data Acquisition** | Months 1 - 2 | Literature review, defining system architecture, ethical clearances, field data collection, and dataset annotation. | Finalized Proposal & Initial Datasets Completed. |
| **Phase 2: Core Algorithm Development** | Months 3 - 5 | Individual model training (Vision, Geospatial, Forecasting, BCS), hyperparameter tuning, and local module testing. | Core Models Trained & Baseline Accuracy Achieved. |
| **Phase 3: System Integration & XAI** | Months 6 - 7 | Establishing cloud API connections, database synchronization, implementing explainability features. | Integrated ADRS Prototype Ready. |

| | | | |
|---|---|---|---|
| **Phase 4: Field Testing & Validation** | Month 8 | System deployment on standard devices, live testing with veterinary officers, performance metric evaluation. | Validation Report & System Refinements. |
| **Phase 5: Documentation & Finalization** | Month 9 | Finalizing the research thesis, formatting codebase, publishing user manuals, and presentation prep. | Final Research Document & System Handover. |

# 9. GANTT CHART




| Title | Start Time | End Time |
|---|---|---|
| **Project Proposal Phase** | | |
| Project Proposal Prep | 2/23 | 3/15 |
| Proposal Submission | 3/15 | 3/15 |
| Proposal Presentation | 3/16 | 3/18 |
| **Implementation & Progress Phase** | | |
| Model Training (New Task) | 3/20 | 4/30 |
| Front End Implementation (New Task) | 4/1 | 6/30 |
| Back End Implementation (New Task) | 4/1 | 6/30 |
| Progress Presentation 1 | 5/4 | 5/6 |
| Interim Task / Ongoing Development | 7/1 | 8/16 |
| Progress Presentation 2 | 8/17 | 8/19 |
| **Finalization & Submission Phase** | | |
| Final Report Preparation | 9/1 | 10/11 |
| Website Development | 10/11 | 10/11 |
| Research Paper Preparation | 9/1 | 10/11 |
| Final Presentation Preparation | 10/11 | 10/11 |
| Final Presentation and Viva | 9/28 | 10/12 |
| Website Evaluation & Logbook Sub | 10/12 | 10/14 |
| Final Report Submission | 9/20 | 10/15 |
| Website Submission | 9/20 | 10/15 |
| Research Paper Submission | 10/15 | 10/15 |
| **Post-Project Phase** | | |
| Publication Evidence Prep | 11/1 | 12/1 |
| Publication Evidence Submission | 12/1 | 12/1 |

*Figure 8: Gantt chart*

# 10. REFERENCES LIST


[1] D. Farrington, N. Andrews, A. Beale, and M. Catchpole, "A statistical algorithm for the early detection of outbreaks of infectious disease," *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, vol. 159, no. 3, pp. 547–563, 1996.

[2] L. Hutwagner, W. Thompson, G. Seeman, and T. Treadwell, "The bioterrorism preparedness and response Early Aberration Reporting System (EARS)," *Journal of Urban Health*, vol. 80, no. 2, pp. i89–i96, 2003.

[3] J. Buckeridge, H. Burkom, M. Campbell, W. Hogan, and A. Moore, "Algorithms for rapid outbreak detection: a research synthesis," *Journal of Biomedical Informatics*, vol. 38, no. 2, pp. 99–113, 2005.

[4] J. Min and S. Agresti, "Modeling infectious disease counts with excess zeros using hurdle and zero-inflated models," *Statistics in Medicine*, vol. 24, no. 15, pp. 2475–2492, 2005.

[5] X. Xu, "A comparison between Markov-switching zero-inflated and hurdle models for infectious disease data," *Statistics in Medicine*, vol. 44, no. 1, pp. 1–15, 2025.

[6] M. Jalali, M. Ahmadi, and A. Bokaie, "Prediction of province-level outbreaks of foot-and-mouth disease using a zero-inflated negative binomial model," *Preventive Veterinary Medicine*, vol. 152, pp. 62–68, 2018.

[7] R. Singh, S. Singh, and R. Kumar, "Trends and future prediction of livestock disease outbreaks using periodic regression analysis," *Veterinary World*, vol. 12, no. 5, pp. 654–660, 2019.

[8] M. S. Islam, M. Rahman, and M. Hasan, "Lumpy skin disease prediction based on meteorological and geospatial features using a random forest algorithm," *Computers and Electronics in Agriculture*, vol. 196, pp. 106905, 2022.

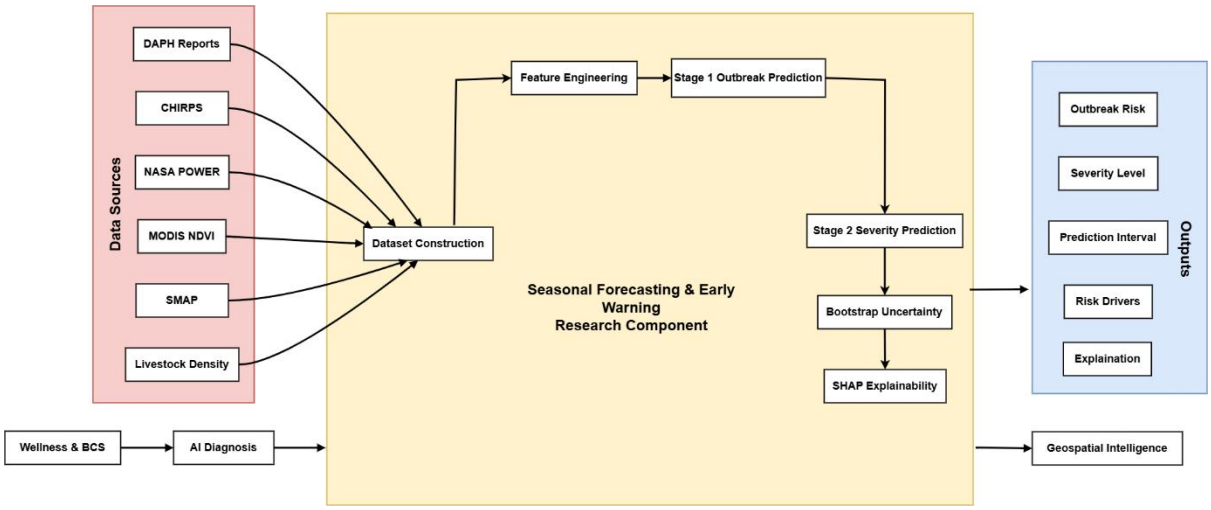
[9] S. Sharma, A. Gupta, and R. Mehta, "Predicting and analysing lumpy skin disease using ensemble machine learning models," *International Journal of Advanced Computer Science and Applications*, vol. 14, no. 3, pp. 1–9, 2023.

[10] A. Kaur and D. Kaur, "Cattle infectious disease prediction using machine learning models and explainable artificial intelligence," *IEEE Access*, vol. 11, pp. 45678–45689, 2023.

[11] Department of Animal Production and Health, "Annual Reports and Veterinary Epidemiology Bulletins," Available: https://daph.gov.lk/services/Ann_Rep

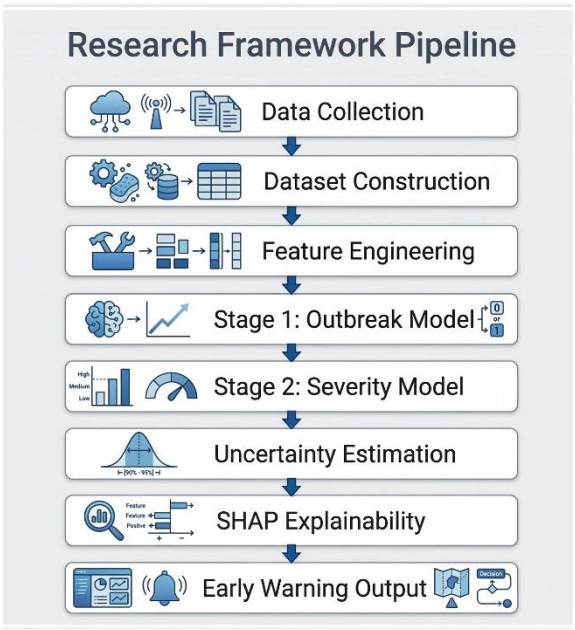
# 11. APPENDICES

Appendix A: Detailed System Architecture Diagram



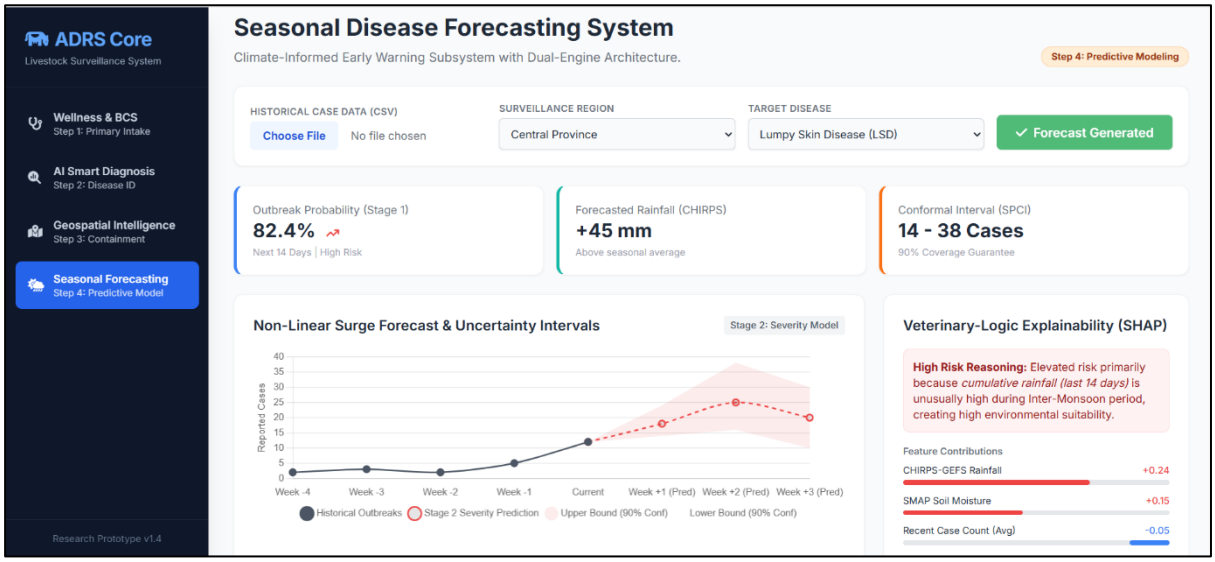

*1: Detailed System Architecture of the Climate-Informed Livestock Disease Forecasting System*

Appendix B: Research Framework Diagram




*2: Research Framework diagram*

Appendix C: Preliminary UI Mockups




*3: Preliminary Dashboard Mockup for Outbreak Risk Monitoring*

Appendix D: Risk Analysis

| Risk | Impact | Mitigation |
|---|---|---|
| Sparse surveillance data | model performance issues | use two-stage modelling |
| Missing environmental data | incomplete features | use interpolation methods |
| Model overfitting | poor generalization | cross-validation and regularization |
| Data delays from DAPH | outdated predictions | periodic data updates |

*4: Risk Analysis*

Appendix E: Ethical Considerations

- The research uses publicly available datasets from DAPH and satellite environmental sources.
- No personal or sensitive data is used.
- Data will be stored securely and used only for research purposes.
- Proper attribution will be given to all data providers.